

Auteur: Yaman Haj Ahmad
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druk de volgende uitdrukking uit in x:

$$\cos(\text{Bgtan}(x))$$

Neem $y = \text{Bgtan}(x)$ waar $y \in \left]-\frac{\pi}{2}, \frac{\pi}{2}\right[$

$$\Leftrightarrow \tan(y) = x$$

$$\Leftrightarrow \frac{\sin(y)}{\cos(y)} = x$$

$$\Leftrightarrow \sin(y) = \cos(y) * x$$

$$\Leftrightarrow \sin^2(y) = \cos^2(y) * x^2$$

$$\Leftrightarrow 1 - \cos^2(y) = \cos^2(y) * x^2$$

$$\Leftrightarrow \frac{1}{\cos^2(y)} - 1 = x^2$$

$$\Leftrightarrow \frac{1}{\cos^2(y)} = x^2 + 1$$

$$\Leftrightarrow \frac{1}{\cos(y)} = \sqrt{x^2 + 1}$$

$$\Leftrightarrow \cos(y) = \frac{1}{\sqrt{x^2 + 1}}$$

$$\Leftrightarrow \cos(\text{Bgtan}(x)) = \frac{1}{\sqrt{x^2 + 1}}$$

References